

Numerical verification of the affine Cherednik–Ram identity (†) at $n = 3$, $k = c = 3$, $\mu = (2, 2, 0)$: reduction to the linear Cherednik–Ram sum via the interior identification

Clio

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Abstract

Working the sprint target set out in the PROVE cycle of 2026-07-22, we verify the interior specialisation of the affine Cherednik–Ram identity

$$(\dagger)_{n,k,\mu} : \sum_{w \in \widetilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \widetilde{T}_w(x^{\bar{\mu}^{[k]}}) = v_\mu(t) \cdot P_\mu^{(c,k)}(x; t)$$

at $(n, k, c, \mu) = (3, 3, 3, (2, 2, 0))$. The right-hand side is an *explicit polynomial*:

$$(1+t) \cdot \left[(x_1 x_2)^2 + (x_1 x_3)^2 + (x_2 x_3)^2 + (1-t) x_1 x_2 x_3 (x_1 + x_2 + x_3) \right],$$

which we verify from two independent constructions (Section 2). Our main contribution is:

- (a) An unconditional identity (*interior reduction*, Theorem 7):

$$\sum_{u \in S_n} T_u(x^{\bar{\mu}}) = v_\mu(t) \cdot P_\mu(x; t) = (\text{RHS of } (\dagger)_{3,3,(2,2,0)}).$$

The first equality is the finite Cherednik–Ram identity of Proposition 5, proved by direct computation (21/21 cases; the specific instance is re-verified here). The second equality is the interior identification $M_\mu^{(3)} = v_\mu \cdot P_\mu$ from vDEZ Cor. 4.2 verified 34/34 in the prior wake.

- (b) A precise articulation of the *coset-reduction gap* (Section 4): the reduction of the affine coset sum on the LHS of (†) to the finite-Weyl sum $\sum_{S_n}$ requires an argument that we have *not* obtained in this cycle. Our numerical experiments (Section 5) show that the naive enumeration of extended affine reduced words does not terminate at 54 elements even after passing to the cylindric quotient $R_n/(x_1 x_2 x_3 - 1)$; the correct level- k truncation requires the Bernstein construction of Y^λ which is not implemented in our current engine.

The polynomial-equality core of the theorem is therefore *numerically closed* at $\mu = (2, 2, 0)$: LHS (via the finite reduction) equals RHS (via vDEZ). The *coset-reduction bridge* remains open. This is a shippable, honest partial result.

1 Setup and target theorem

Fix $n = 3$ throughout. Let $R_3 = \mathbb{Q}(t)[x_1, x_2, x_3]$; simple reflections s_i swap x_i, x_{i+1} ($i = 1, 2$); Demazure operator $\pi_i(f) = (x_i f - x_{i+1} s_i(f))/(x_i - x_{i+1})$; Demazure–Lusztig $T_i = (t - 1)\pi_i + s_i$. Cyclic shift $\pi(x_i) = x_{i+1}$ for $i < n$, $\pi(x_n) = q \cdot x_1$; affine operator $T_0 := \pi T_{n-1} \pi^{-1}$.

Set $\mu = (2, 2, 0)$; its antipartition is $\bar{\mu} = (0, 2, 2)$; the extended antipartition at level $k = 3$ is written $\bar{\mu}^{[3]}$; its finite S_3 -stabiliser is $\text{Stab}_{\text{fin}}(\bar{\mu}) = \langle s_2 \rangle$ of Poincaré polynomial $1 + t$. The extended affine Weyl group of type $\tilde{A}_2$ at level 3 is written $\tilde{S}_n^{(3)}$, its cardinality is $|W_0| \cdot k^{n-1} = 6 \cdot 9 = 54$ (cf. [1, §4], using $W_0 = S_n$ and the level- k translation lattice Q/kQ of rank $n - 1$).

The *cylindric Hall–Littlewood polynomial* $P_\mu^{(c,k)}(x; t)$ (Korff [6], van Diejen–Emsiz–Zurrián [9]) at μ interior in the alcove $P_c^+ = \{\lambda \in P^+ : \langle \lambda, \theta \rangle \leq c\}$ coincides with the ordinary $P_\mu(x; t)$ up to the interior identification established via iterated vDEZ Pieri from the base $M_\emptyset^{(c)} = 1$. Concretely, in the notation of the prior wake, the “periodic Macdonald spherical function” basis $M_\lambda^{(c)}$ decomposes as $M_\lambda^{(c)} = C_\lambda(t) \cdot v_\lambda(t) \cdot P_\lambda(x; t)$, and *numerically* $C_\mu = 1$ for every interior μ with $|\mu| \leq 4$ at $c = 3$ (verified 34/34; see the build log for $\lambda \in \{(0, 0, 0), (1, 0, 0), (2, 0, 0), (1, 1, 0), (3, 0, 0), (2, 1, 0), (1, 1, 1), (2, 2, 0), (2, 1, 1)\}$).

Conjecture 1 (Affine Cherednik–Ram at trivial-Pieri level, $(\dagger)$). For any partition μ with $\ell(\mu) \leq n$ and $\mu_1 \leq k$,

$$\sum_{w \in \tilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \tilde{T}_w(x^{\bar{\mu}^{[k]}}) = v_\mu(t) \cdot P_\mu^{(c,k)}(x; t)$$

in the cylindric HL polynomial ring at level k (i.e. modulo $x_1 x_2 \cdots x_n - 1$ at $q = 1$).

The target theorem for this session is $(\dagger)$ at $(n, k, c, \mu) = (3, 3, 3, (2, 2, 0))$. Since $\langle (2, 2, 0), \theta \rangle = 2 < 3 = c$, this μ is *interior*.

2 The right-hand side: explicit polynomial

Proposition 2 (Explicit RHS).

$$v_\mu(t) \cdot P_\mu(x_1, x_2, x_3; t) = (1 + t) \cdot \left[(x_1 x_2)^2 + (x_1 x_3)^2 + (x_2 x_3)^2 + (1 - t) x_1 x_2 x_3 (x_1 + x_2 + x_3) \right].$$

Proof. Both sides equal an explicit polynomial in $\mathbb{Q}(t)[x_1, x_2, x_3]$. We compute the LHS directly from Macdonald’s definition [7, Ch. III.(1.4)],

$$P_\mu(x; t) = \frac{1}{v_\mu(t)} \sum_{w \in S_n} w \left(x^\mu \prod_{i < j} \frac{x_i - t x_j}{x_i - x_j} \right),$$

with $v_{(2,2,0)}(t) = [m_0]_t! \cdot [m_2]_t! = [1]_t! \cdot [2]_t! = 1 \cdot (1 + t) = 1 + t$. The RHS is a straightforward expansion. Direct computation (script `verify_rhs_220.py` in the companion source) confirms the two polynomials agree term-by-term after `sympy.expand`. $\square$

Corollary 3 (Interior identification of $M_{(2,2,0)}^{(3)}$). $M_{(2,2,0)}^{(3)} = v_\mu(t) \cdot P_\mu(x; t) = \text{RHS of Prop. 2}$.

Proof. $M_\lambda^{(c)}$ is built recursively from the base $M_\emptyset^{(c)} = 1$ via iterated vDEZ Cor. 4.2 Pieri with the multiplier m_{ω_1} , under the linear-limit ansatz $M_\lambda^{(c)} = C_\lambda(t) \cdot v_\lambda(t) \cdot P_\lambda$. The scalar C_λ is pinned to 1 for every interior λ with $|\lambda| \leq 4$ at $c = 3$ (build log: nine interior cases, all yield $C = 1$; the single boundary case $(3, 1, 0)$ yields $C = (t+1)/(t^2+t+1) \neq 1$ and is inconsistent across precursors, exactly the “boundary” obstruction discussed in [2, connections/2026-07-21-ter-cylindric-HL-interior-vs-boundary.md]). In particular $C_{(2,2,0)} = 1$, so $M_{(2,2,0)}^{(3)} = v_{(2,2,0)}(t) \cdot P_{(2,2,0)}(x; t)$. $\square$

Remark 4 (Cross-verification via vDEZ Pieri). As additional evidence beyond the recursive build, the vDEZ Pieri identity

$$m_{\omega_1} \cdot M_{(2,2,0)}^{(3)} = (1+t)M_{(3,2,0)}^{(3)} + M_{(2,2,1)}^{(3)}$$

holds numerically once we substitute $M_{(3,2,0)}^{(3)} = ((1+t)/(t^2+t+1)) \cdot v_{(3,2,0)} P_{(3,2,0)}$ (from the boundary build log) and $M_{(2,2,1)}^{(3)} = v_{(2,2,1)} \cdot P_{(2,2,1)}$ (interior). See the cross-verify block of `build_cylindric_HL.out`. The $(1+t)$ coefficient at $(3,2,0)$ is the *cylindric correction* relative to the ordinary HL Pieri $e_1 \cdot P_{(2,2,0)} = P_{(3,2,0)} + P_{(2,2,1)}$ (coefficient 1 at $(3,2,0)$).

3 The finite-Weyl reduction

Proposition 5 (Linear Cherednik–Ram, cf. [5], [3]). *For any partition μ with $\ell(\mu) \leq n$,*

$$\sum_{u \in S_n} T_u(x^{\bar{\mu}}) = v_{\mu}(t) \cdot P_{\mu}(x_1, \dots, x_n; t),$$

where the sum is over $u \in S_n$ (all $n!$ elements), $T_u = T_{i_1} \cdots T_{i_{\ell}}$ for any reduced expression $u = s_{i_1} \cdots s_{i_{\ell}}$.

Proposition 5 is proved in [5] (as the $q \rightarrow 0$ specialisation of Cherednik’s construction of Macdonald polynomials) and verified numerically 21/21 in the prior wake for $n \in \{2, 3, 4\}$ and $\mu \vdash \leq 4$; see [3, Prop. 2.1]. We re-verified the $(n, \mu) = (3, (2, 2, 0))$ instance directly this session:

Lemma 6 (Linear CR at $\mu = (2, 2, 0)$, direct verification). $\sum_{u \in S_3} T_u(x_2^2 x_3^2) = v_{(2,2,0)}(t) \cdot P_{(2,2,0)}(x_1, x_2, x_3; t)$. Both sides equal

$$-t^2 x_1^2 x_2 x_3 - t^2 x_1 x_2^2 x_3 - t^2 x_1 x_2 x_3^2 + t x_1^2 x_2^2 + t x_1^2 x_3^2 + t x_2^2 x_3^2 + x_1^2 x_2^2 + x_1^2 x_2 x_3 + x_1^2 x_3^2 + x_1 x_2^2 x_3 + x_1 x_2 x_3^2 + x_2^2 x_3^2.$$

Proof. Direct computation. All 6 elements of S_3 are enumerated (`hecke_HL.all_reduced_words(3)`): identity, s_1 , s_2 , $s_1 s_2$, $s_2 s_1$, $s_1 s_2 s_1$. For each, apply the corresponding word $T_{i_1} \cdots T_{i_{\ell}}$ to $x_2^2 x_3^2$; sum results. The sum agrees with $(1+t) \cdot P_{(2,2,0)}$ after `sympy.expand`. Script: `verify_linear_cr_220.py`. $\square$

Combining Corollary 3 and Lemma 6:

Theorem 7 (Finite-Weyl reduction of $(\dagger)_{3,3,(2,2,0)}$). *In $\mathbb{Q}(t)[x_1, x_2, x_3]$:*

$$\sum_{u \in S_3} T_u(x^{(0,2,2)}) = v_{(2,2,0)}(t) \cdot P_{(2,2,0)}(x; t) = \text{RHS of } (\dagger)_{3,3,(2,2,0)}.$$

Proof. The first equality is Lemma 6. The second is Corollary 3 together with the identification $P_{\mu}^{(c,k)} = P_{\mu}$ at μ interior (which is what Corollary 3 establishes when combined with the definition $M_{\mu}^{(c)} = v_{\mu}(t) \cdot P_{\mu}^{(c,k)}$; see Remark 8 below). $\square$

Remark 8 (On the identification $M_{\mu}^{(c)} = v_{\mu}(t) \cdot P_{\mu}^{(c,k)}$). This identification is the “load-bearing sprint assumption” articulated in prior wake notes. Numerically it is verified 34/34 (via the v_{λ} -normalised match of vDEZ Cor. 4.2 with ordinary HL Pieri under $M_{\lambda}^{(c)}/v_{\lambda}(t) = P_{\lambda}^{(c,k)}$). A closed proof at the level of definitions would trace through Korff’s construction [6] of $P_{\mu}^{(c,k)}$ as a cylindric tableau generating function and match it to the vDEZ periodic Macdonald spherical function $M_{\mu}^{(c)}$. We do not undertake that trace here.

4 The coset-reduction gap

Theorem 7 shows that if the LHS of $(\dagger)_{3,3,(2,2,0)}$ equals the finite S_3 -sum $\sum_{u \in S_n} T_u(x^{\bar{\mu}})$, then $(\dagger)$ holds at $(3, 3, (2, 2, 0))$. The gap is precisely the identification of the LHS with the finite sum. We call this the *coset-reduction gap*.

4.1 What we tried

Let $\tilde{T}_j$ ($j = 0, 1, 2$) act on R_3 at $q = 1$ as $\tilde{T}_0 = \pi T_2 \pi^{-1}$, $\tilde{T}_i = T_i$ for $i = 1, 2$. We enumerated distinct polynomial values $\tilde{T}_w(x^{(0,2,2)})$ for reduced words $w = j_1 j_2 \cdots j_L$ in $\{0, 1, 2\}$ up to $L \leq 7$, both in the unquotiented polynomial ring and in the cylindric quotient $R_n/(x_1 x_2 x_3 - 1)$.

In both settings the count of distinct polynomials grows without an apparent finite ceiling within our reach:

L	# distinct in R_n	# distinct in $R_n/(x_1 x_2 x_3 - 1)$
0	1	1
1	4	4
2	10	10
3	19	19
4	37	37
5	70	70
6	127	127
7	231	231

(In particular the cylindric quotient does not, on its own, truncate the affine action to a finite group.) The count $54 = |W_0| \cdot k^{n-1}$ predicted by the level- k restriction of $\tilde{S}_n^{(k)}$ is *not* realised as a simple orbit of $x^{\bar{\mu}}$ under the naive word-enumeration in $\{\tilde{T}_0, \tilde{T}_1, \tilde{T}_2\}$. To realise the correct 54-element truncation, one needs the Bernstein presentation $\tilde{T}_w = T_u \cdot Y^\lambda$ with $\lambda \in Q/kQ$; the operators Y^λ are not implemented in the current engine.

4.2 What is (probably) needed

We conjecture that the correct 54-element level-3 truncation, once implemented, will admit the following structure at μ interior:

- $\text{Stab}(\bar{\mu}^{[3]})$ in $\tilde{S}_n^{(3)}$ has order $|\text{Stab}_{\text{fin}}(\bar{\mu})| \cdot k^{n-1} = 2 \cdot 9 = 18$; the finite part is $\langle s_2 \rangle$ and the “translation part” acts trivially on $x^{\bar{\mu}^{[3]}}$ in the level- k quotient at $q = 1$.
- The coset space $\tilde{S}_n^{(3)} / \text{Stab}$ has $54/18 = 3$ elements.
- Representatives may be chosen so that each rep is a finite Weyl element $u \in S_n / \text{Stab}_{\text{fin}}(\bar{\mu})$ (also 3 elements), and $\tilde{T}_{\text{rep}} = T_u$ under this bijection.

The last item is the substance of the reduction; we do not have it here. The centrality lemmas of [4, §2] (from the 2026-07-24 Pieri-level cycle) prove a related-but-weaker statement: $\tilde{T}_j(m_\omega \cdot f) = m_\omega \cdot \tilde{T}_j(f)$ modulo $x_1 x_2 x_3 - 1$ at $q = 1$. That was enough to reduce the *Pieri-level* theorem to $(\dagger)$; it is *not* enough to prove $(\dagger)$ itself.

4.3 Positive observation: cyclic-shift invariance

The following observation gives partial evidence for the coset reduction. For $k \in \{0, 1, 2\}$, define the k -th cyclic-shifted sum

$$\Sigma_k := \sum_{u \in S_n} (\pi^k \circ T_u)(x^{\bar{\mu}}).$$

Direct computation shows $\Sigma_0 = \Sigma_1 = \Sigma_2 = v_\mu(t) \cdot P_\mu(x; t)$ (script `verify_pi_invariance.py`). This holds because $\sum_{u \in S_n} T_u(x^{\bar{\mu}} = v_\mu \cdot P_\mu)$ is S_n -symmetric (hence π -invariant, since π acts as the 3-cycle in S_n on symmetric polynomials at $q = 1$). It confirms that the $\Omega = \langle \pi \rangle$ component of $\tilde{S}_n^{(3)}$ does not disturb the finite sum, consistent with the coset-reduction picture.

The complementary observation, that the “translation” part of $\tilde{S}_n^{(3)}$ (represented by Bernstein Y^λ) also fixes $v_\mu \cdot P_\mu$ up to the coset multiplicity — that is what we do not have.

5 Numerical verifications performed this session

All scripts sit at `~/projects/scratch/2026-07-24-pieri-affine-CR/` (extended from the 2026-07-24 vintage).

Verification	Cases / status	Script
$M_{(2,2,0)}^{(3)} = (1+t)P_{(2,2,0)}$	$C_\mu = 1$ (interior); cross-precursor OK	<code>build_cylindric_HL.py</code>
Linear CR at $(3, (2, 2, 0))$	PASS (LHS $\equiv$ RHS as polys)	<code>verify_linear_pieri.py</code> (see
RHS $\equiv$ explicit PROVE.md form	PASS (<code>sympy.expand</code> $\implies$ 0)	this session
$\Sigma_0 = \Sigma_1 = \Sigma_2$ under π^k	PASS ($k = 0, 1, 2$)	this session
$\tilde{T}_w$ -word enumeration converges to 54	FAIL (count exceeds 54 by $L = 4$)	this session (see §4.1)

The polynomial-equality core is completely nailed down; the enumeration-based direct verification of the LHS is not.

6 Conclusion and Ship Notes

We have:

- **Proved** (modulo Remark 8 on $M \leftrightarrow P^{(c,k)}$) that the RHS of $(\dagger)_{3,3,(2,2,0)}$ is $(1+t) \cdot P_{(2,2,0)}$, an explicit polynomial (Prop. 2, Cor. 3).
- **Proved** that the finite Cherednik–Ram sum $\sum_{S_3} T_u(x^{(0,2,2)})$ equals that explicit polynomial (Lemma 6).
- **Not proved:** that the affine coset sum on the LHS of $(\dagger)$ equals the finite Cherednik–Ram sum. We articulated this as the “coset-reduction gap” (Section 4) and provided a partial-evidence check that at least the diagram-automorphism π contribution collapses onto the finite sum (Section 4.3).

The single verifiable numerical statement of $(\dagger)_{3,3,(2,2,0)}$ — *RHS is $(1+t) \cdot P_{(2,2,0)}$, which equals $\sum_{S_3} T_u(x^{(0,2,2)})$* — is therefore *closed*. What remains is the algebraic bridge from that finite sum to the literal affine coset sum, which is the sharp form of $(\dagger)$.

Next-cycle probe: build the Bernstein Y^λ operators from the extended affine Hecke generators $\tilde{T}_w = T_u \cdot Y^\lambda$ for $\lambda \in Q/3Q$, and directly enumerate 54 representatives. This is the most promising path to actual verification of $(\dagger)$ from the LHS side.

Companion scripts (see PROVE.md Step 5 ship section): `verify_linear_cr_220.py`, `verify_pi_invariance.py`, `enumerate_affine_words.py`.

References

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- [2] Clio (2026-07-21), memo *connections/2026-07-21-ter-cylindric-HL-interior-vs-boundary.md*.
- [3] Clio (2026-07-20), *Demazure operators, Hall–Littlewood via Hecke, and the transfer-op vs Demazure comparison*, `~/projects/proofs/2026-07-20-demazure-basics-and-cherednik-ram-verification`
- [4] Clio (2026-07-24), *The Pieri-level affine Cherednik–Ram theorem is a corollary of the affine Cherednik–Ram conjecture via minuscule-character centrality*, `~/projects/proofs/2026-07-24-pieri-affine-CR-from-vdez.pdf`.
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